

Haken Continuum Interfacing Cookbook

Interfacing Control Devices and Third-Party Software to
Haken Continuum, ContinuuMini and EaganMatrix Module
Firmware versions 10.x and later.

Hardware Interfacing to Continuum Devices

Arturia BeatStep and BeatStep Pro Configuration

Step 1. Bring up Arturia Midi Control Center (make sure Haken Editor is not running)



Step 2. For Rotaries to set a CC for Continuum control, set Mode: Control, Option: Relative #1, Channel: 2, CC: Desired CC (see Continuum Manual). Here setting Macro Controller i (cc 12).



Note: When finished configuring your BeatStep or Beatstep Pro, save your preset in the Control Center and load into the Beatstep.

In 9.9x and later firmware, Channel 2 must be used when running Beatstep through Haken Editor. This will work without editor as well. Use a 3.5mm TS cable to connect Midi port to EaganMatrix module or Slim Continuum.

Step 3. To set one of the 16 pads to bring up one of the 16 user presets. Mode: Patch Change, Channel: 2, Program Number 0 = usr01, 1=usr02, ... 15 = usr16 (example below), Bank LSB: 0, Bank MSB: 0.



Step 4. To set one of the 16 pads to bring up a User preset past usr16 (up to user preset usr128): Patch Change, Channel: 2, Program Number 16-127 as follows:

Bank 1 Presets 1-16, PC = 0..15
 Bank 2 Presets 17 - 32, PC = 16..31
 Bank 3 Presets 33 - 48, PC = 32..47
 Bank 4 Presets 49 - 64, PC = 48..63
 Bank 5 Presets 65 - 80, PC = 64..79
 Bank 6 Presets 81 - 96, PC = 80..95
 Bank 7 Presets 97 - 112, PC = 96..111
 Bank 8 Presets 113 - 128, PC = 112..127

For example, the following programs Pad15 when pressed to bring up preset "usr65".



Step 5. To set one of the 16 pads to bring up one of the 500+ System Presets, you will enter a Preset Category in Bank MSB and then the preset within that category with Program Number. Note: Category 9 does not apply to ContinuumMini which only uses 1-12, renumbered removing PR (Slim 10/Mini 9, etc.)

LSB is left at 0 (but will be used if there are more than 128 presets in a category).

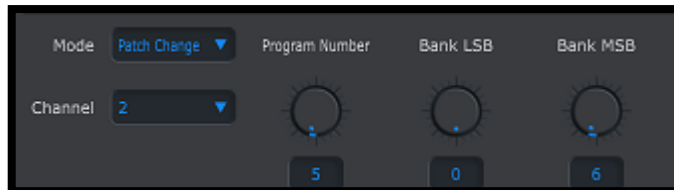
Bank MSB Categories:

- 1 = **ST** (Strings)
- 2 = **WI** (Winds)
- 3 = **VO** (Vocal)
- 4 = **KY** (Keyboard)
- 5 = **CL** (Classic)
- 6 = **OT** (Other)
- 7 = **PE** (Percussion)
- 8 = **PT** (Tuned Percussion)
- 9 = **PR** (Processor)
- 10/9 = **DO** (Drone)
- 11/10 = **MD** (Midi)
- 12/11 = **CV** (CVC)
- 13/12 = **UT** (Utility)

Here is the 1st preset in the ST Category (ST 01)



Here is the 6th preset in the OT Category (OT 06)



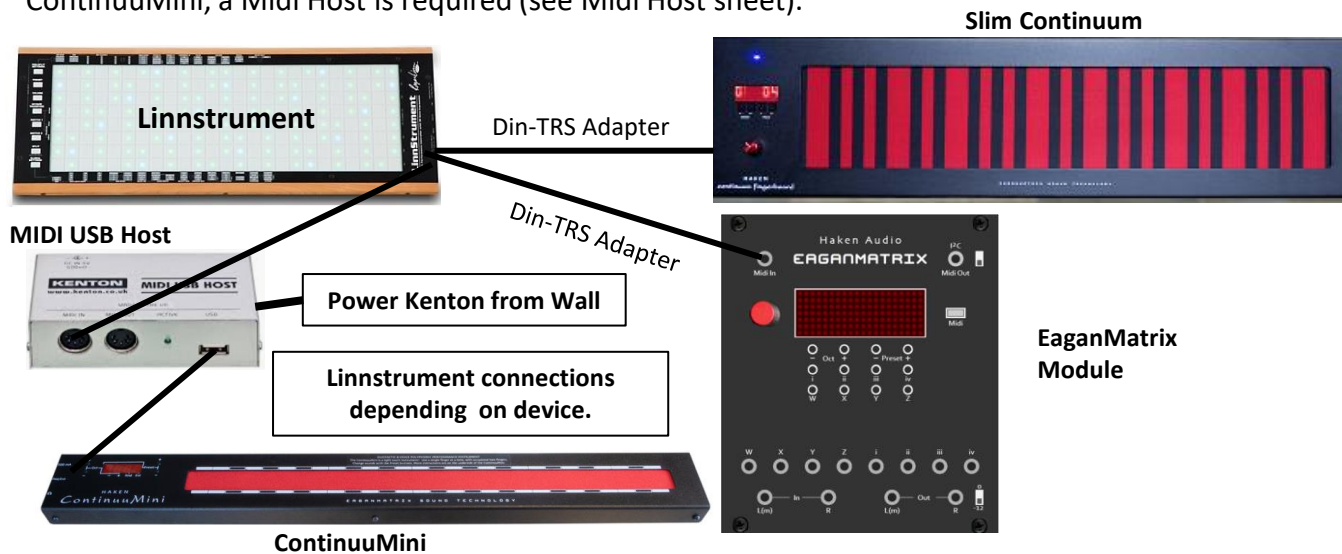
Here is the 128th preset in the OT Category (OT 128)

Note LSB is set to 1, MSB=6. Set PC=1 for presets OT129, etc.



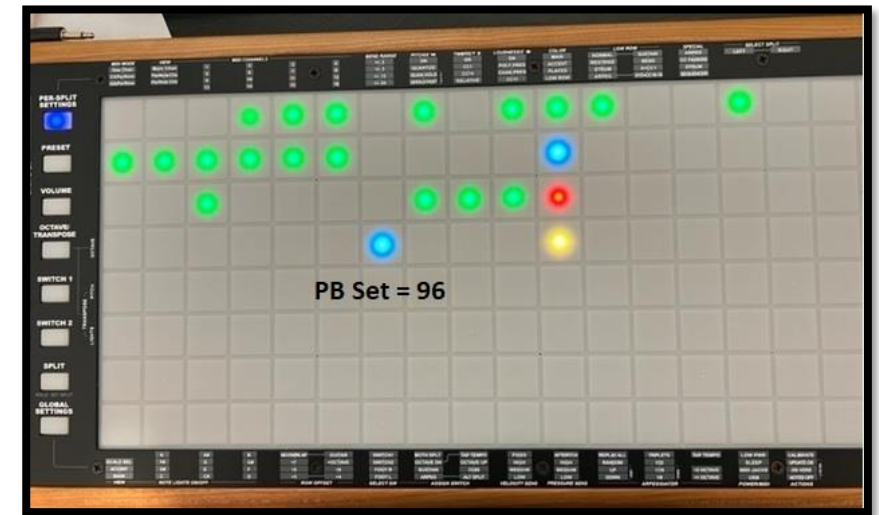
Using Linnstrument with Continuum MPE Devices

Step 1. For connection without Computer: Connect Linnstrument Midi DIN Out to Half/Full size Continuum Midi DIN In or Slim/EaganMatrix Module (using DIN-TRS adapter) Midi In. For ContinuuMini, a Midi Host is required (see Midi Host sheet).



Note: Some Kenton hosts do not put out required 5V to adequately power ContinuuMini. See hosts page for other options.

Step 3. Configure Linnstrument and possibly configure Preset in Haken Editor to match desired setup.

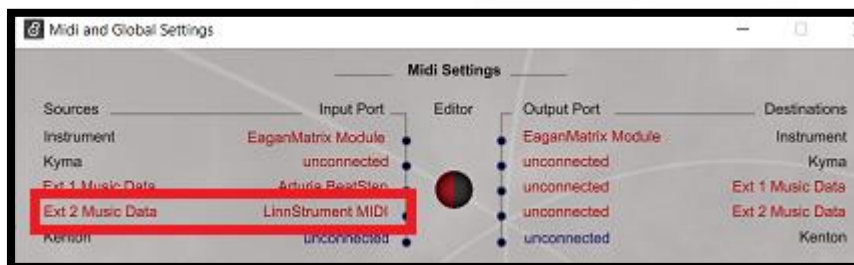
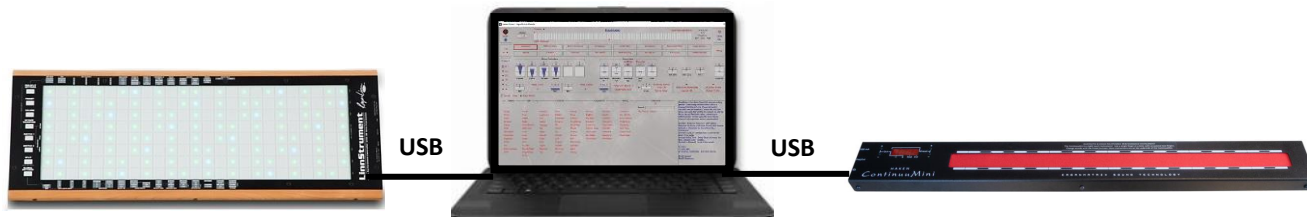


On Linnstrument:

- Set Midi Mode = ChPerNote
- Set View = PerNote Chs
- Set Midi Channels = 2..Max Polyphony+1 (as set in your desired EaganMatrix preset)
- Set Pitch = On
- Set Timbre=C74
- Set Loudness/Z = Chan Pres (Channel Pressure)
- Set Bend Range = 12, 24, 36, 48 or 96 (Default for most presets is 96)

On Continuum/ContinuuMini:

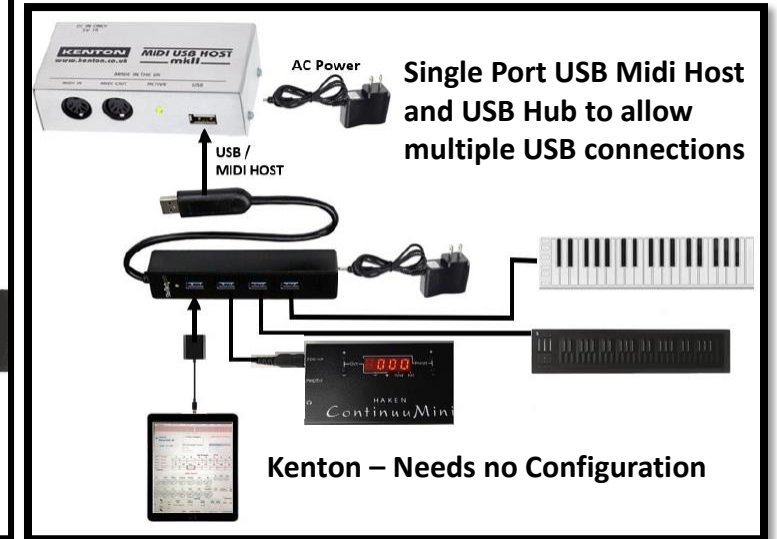
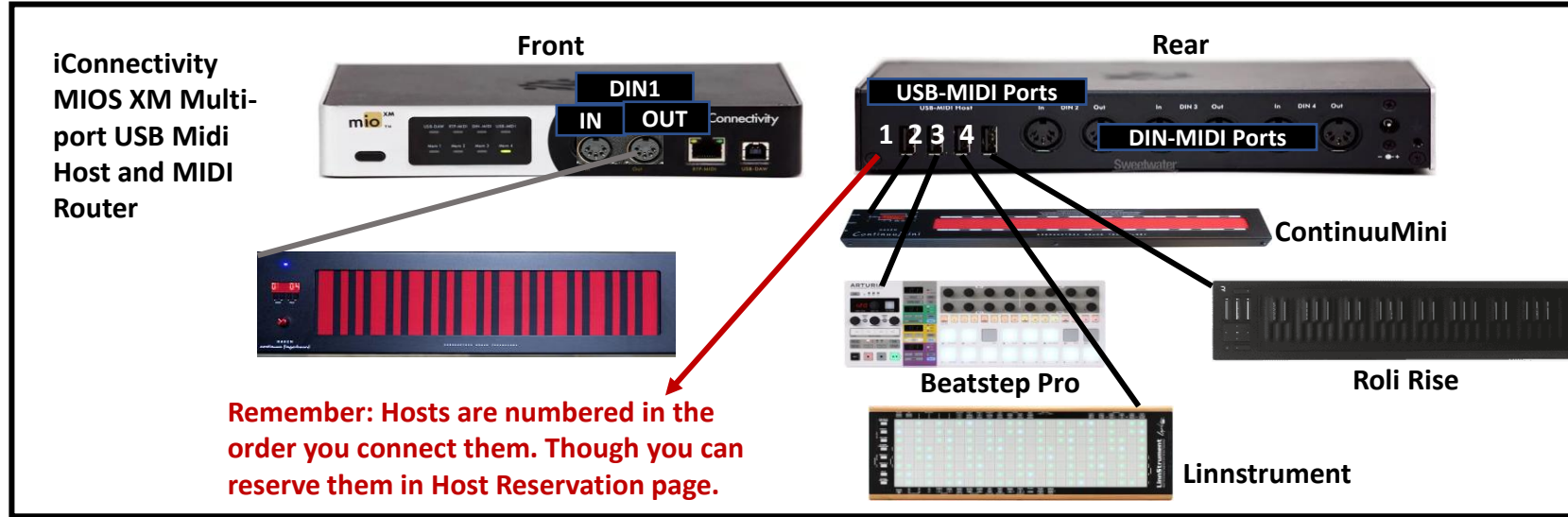
- Set Y/Z= "MPE" for desired preset ("MPE+" will work too)
- Set Z/Bend = 12, 24, 36, 48 or 96 (match with Linnstrument)



Step 2. If playing through Computer/ Haken Editor Select the "Ext 1" or "Ext 2" input as Linnstrument Midi USB device (Linnstrument must be connected before you bring up the editor)

Connecting USB Midi Controllers Directly to ContinuuMini/EaganMatrix Module with Multi-port iConnectivity MIO XM Midi Host or Single Port host and USB Hub (1)

In the case of the MIO-XM or MIO-XL that have multiple USB MIDI Host ports capable of outputting 500ma, you don't need the USB hub (frankly this is a more reliable and simple option). If using Continuum, not ContinuuMini, connect MIDI DIN Ports of the interface to the Continuum. Configure the USB MIDI Host to accept input and output on the desired ports with "Auracle for X-Series" software app. Note: MIO will automatically detect which port USB port is connected in what order and make port assignments so note connection sequence.



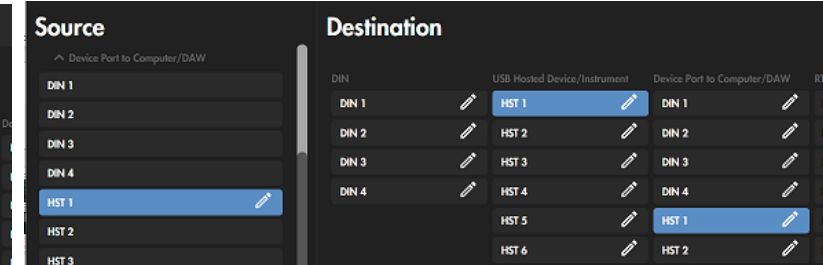
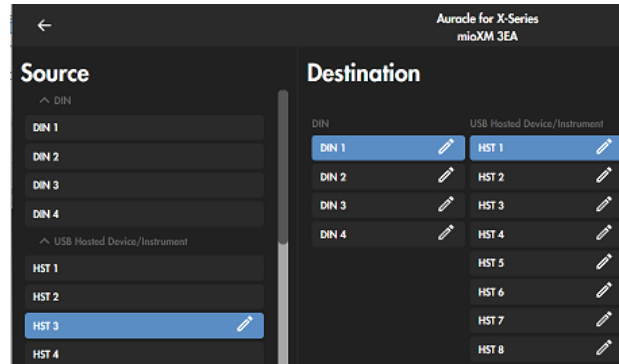
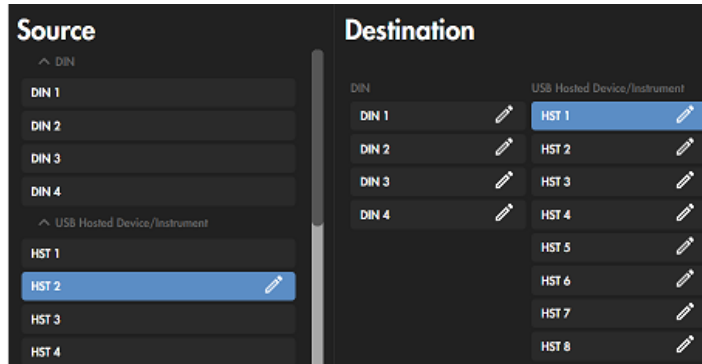
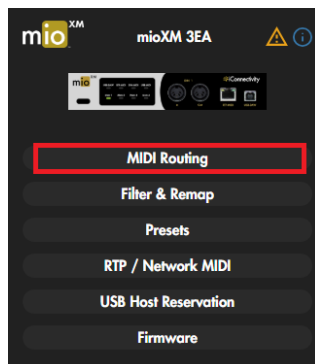
MIO XM Configuration (Assume ContinuuMini on HST/Port1, Arturia BeatStep on HST/Port 2, Linnstrument on HST/Port 3 and Roli Rise on HST/Port 4)

Step 1: Bring up Auricle Configuration App

Step 2: Configure what Source host ports connected should send data to what Destination USB host ports. Here BeatStep Pro HST 2 Source sends to ContinuuMini (HST1).

Step 3: Add other connections. Here Linnstrument (HST3) sends data to ContinuuMini (HST1) and also to the Slim Continuum connected to DIN1.

Step 4: If using DAW add connections for which Host ports will connect to DAW and output to DAW destination ports and any host devices. Not needed if not using a computer for Continuum device Midi Host port connections.



Note: You can see its possible with MIO to route anything to anything else plus it also supports RTP Midi over Ethernet routing as well (see MIO XM documentation and online videos)

Connecting USB Midi Hosts to Directly to ContinuuMini/Continuum without Computer (2)

You cannot connect a USB MIDI controller directly to the ContinuuMini or USB connection with Slim Continuum or EaganMatrix Module (though Slim and EM Module have MIDI TRS connections). You will need a **USB MIDI Host** in the connection path to do this (though MIDI devices can directly play the Continuum through the MIDI DIN ports (or 3.5mm TS/TRS MIDI ports in the case of the “Slim” model and EM Module). **Note:** Your MIDI Host must put out enough power (500ma) and voltage (5V) to power the ContinuuMini.

- iConnectivity MIO XM (4 USB MIDI Host ports, 4 sets MIDI DIN, 1 RTP MIDI, 1 USB DAW Computer Connection)



- iConnectivity MIO XL (Rack mount) – (10 USB MIDI Host ports, 10 sets MIDI DIN, 1 RTP MIDI, 1 USB DAW connection)



- **Kenton MKII/III USB MIDI Host** (1 USB MIDI Host, 1 set MIDI DIN – Must be powered from wall for reliable Continuum connection. Nothing to configure.



Note: Some have had issues (especially with new MK3 version) with Kenton Midi Host not delivering a steady 5V required by ContinuuMini.

- **Raspberry PI** – Needs ALSA MIDI Software

To Configure as Midi Host:

1. Connect keyboard, mouse, monitor & boot
2. Install ALSAMidi package if not installed (alsa.opensrc.org)
3. Connect USB Midi Devices and reboot
4. Type “aconnect -l” to verify also installed, in not look up how
5. Type “aconnect -i” to see Midi Client devices and device numbers
6. Type “connect nn nn” to make Midi client Host connections (here “connect 14 16” to connect NanoKey to ContinuuMini)
7. Disconnect and you can use standalone
8. Connect same devices to same ports when using



```
pi@raspberrypi1 ~$ aconnect -l
client 0: 'System' [type=kernel]
  0 'Timer'
  1 'Announce'
client 14: 'Midi Through' [type=kernel]
  0 'Midi Through Port-0'
client 16: 'ContinuuMini SN000055' [type=kernel]
  0 'ContinuuMini SN000055 MIDI 1'
client 20: 'nanoKEY' [type=kernel]
  0 'nanoKEY MIDI 1'
```

- **CME WIDI UHost** (wireless) – Must provide 5V power



Others – But not tested and verified

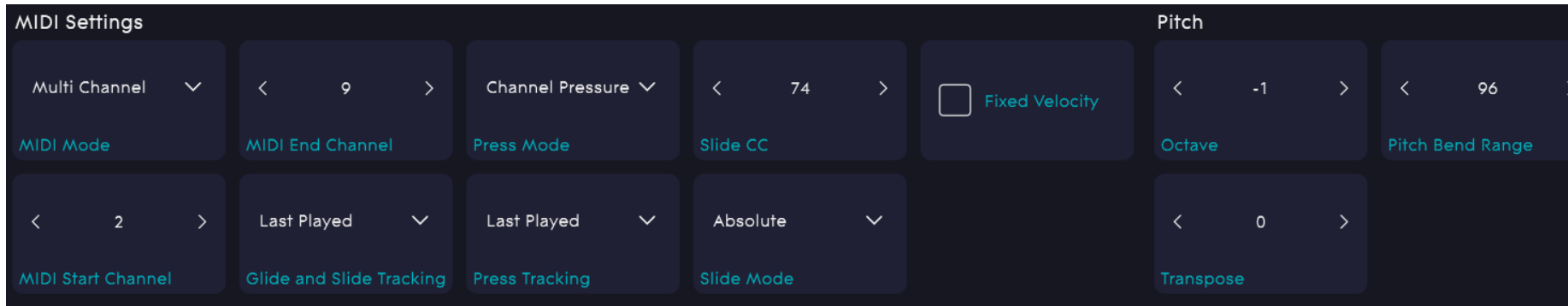
- MiDiPlus USB MIDI Host
- CerebelUSB MIDI Host
- MicroGhost USB MIDI Host
- UHM USB MIDI Host
- MiDiTech USB MIDI Host
- BomeBox
- RK006
- DoReMi (Note: mine did not work)
- Likely numerous others as well

Connecting Roli Rise to Haken Continuum Device

Step 1 – Connect Roli to computer and run Roli Dashboard. Configure as follows (for Continuum MPE connection)

On Computer:

- Bring up Roli Dashboard
- Set MIDI Channel Mode = Multi Channel (not MPE because we need to define starting and ending channels)
- Set Channel Range = 2..Max Polyphony+1 (as set in your desired Mini preset – for example channels 2-9)
 - Set Midi Start Channel = 2
 - Set Midi End Channel = 9
- Set Press Mode = Channel Pressure
- Set Slide CC = 74
- Set Glide and Slice Tracking = Last Note Played (default - or to another option if you choose)
- Set Press Tracking (Z) = Last Note Played (default – or to another option if you choose)
- Set Pitch Bend Range = 96 (or less if you don't need large bend – just match ContinuumMini setting)
- Set Octave setting to match your desired range of ContinuumMini output (normally this would be down one octave on Rise – but adjust as you see fit).
- If using a pedal, set that to desired CC (see CC control section in this manual). In this case make sure EaganMatrix is set to receive “All” Midi data, not just “Music” data (if connecting through EM).



Step 2 - Verify on Continuum Device:

- Set Y/Z= “MPE” for desired preset (“MPE+” will work too)
- Set Z/Bend = 12, 24, 36, 48 or 96 (or whatever you set Roli to – normally you will match Roli to Continuum device)

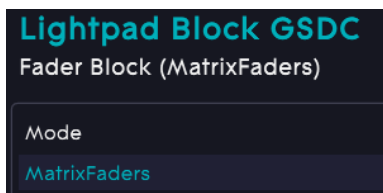
Step 3 – Connect Roli Rise per one of the previous diagrams (Midi Host needed to connect to USB devices without computer or use Ext1/Ext2 connect in Editor).

Connecting Roli Light Block as CC Controller to Haken Continuum Device

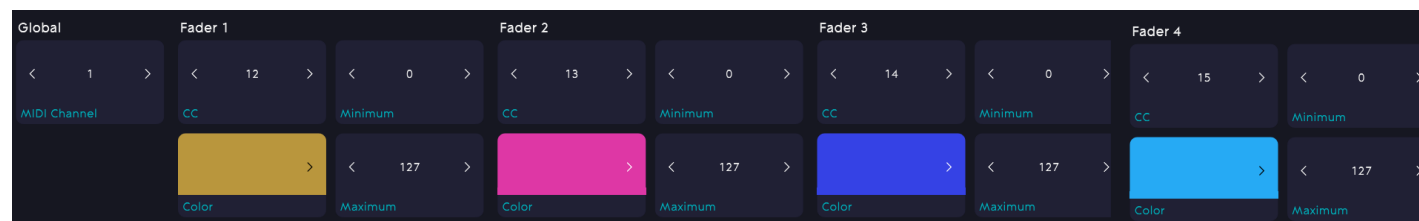
Step 1 – Connect Roli light Block to computer and run Roli Dashboard

Step 2 – Configure for Desired App

For Fader Bank – Select Fader option, Edit

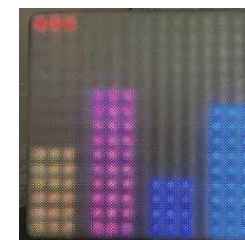


1. You can configure four banks of four faders (Fader1 – Fader 16)
2. Set Global Mode = 1 (Control CCs all work on Channel 1, or Channel 2)
3. Set Fader CC, Fader Minimum (normally 0) and Fader Maximum (normally 127)
4. Set Desired Fader color
5. Repeat for other fader groups as desired
6. Save your configuration (save as default if desired)



Here configured for Macro Controllers (i-CC12, ii-CC13, iii-CC14 and iv-CC15)

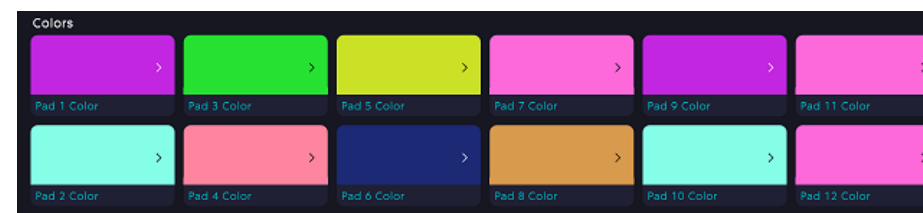
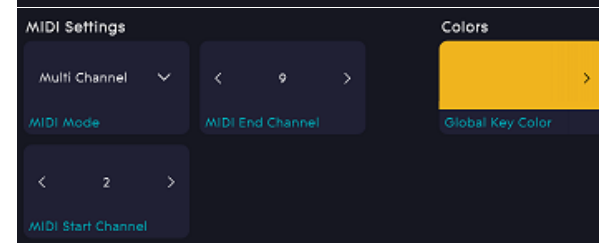
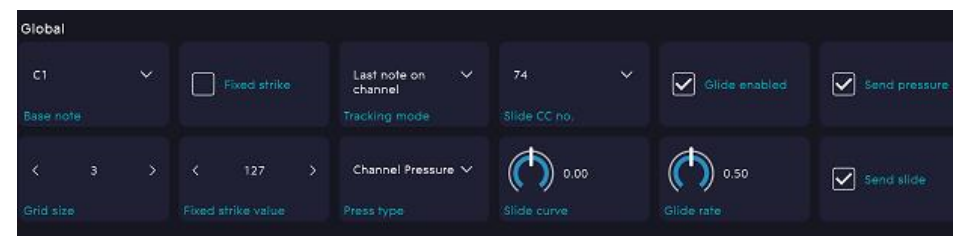
Numerous Apps to configure that might work with Continuum



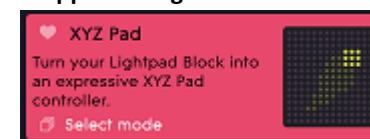
For Drum (MPE) Bank – Select Fader option, Edit (Note: there are other App options)



1. Select Multi Channel mode (i.e. 3x3 Multi-Channel Mode)
2. Set Global Base Note (default C1)
3. Set MPE Mode (default Last note on channel)
4. Set Channel Pressure (default)
5. Set Slide CC = 74 (default)
6. Keep other Global defaults
7. Set Midi Mode = Multi Channel (not MPE)
8. Set Midi Start Channel = 2, Midi End Channel = Preset Polyphony + 1
9. Save your configuration (save as default if desired)



Also check out the XYZ pad App that might be useful

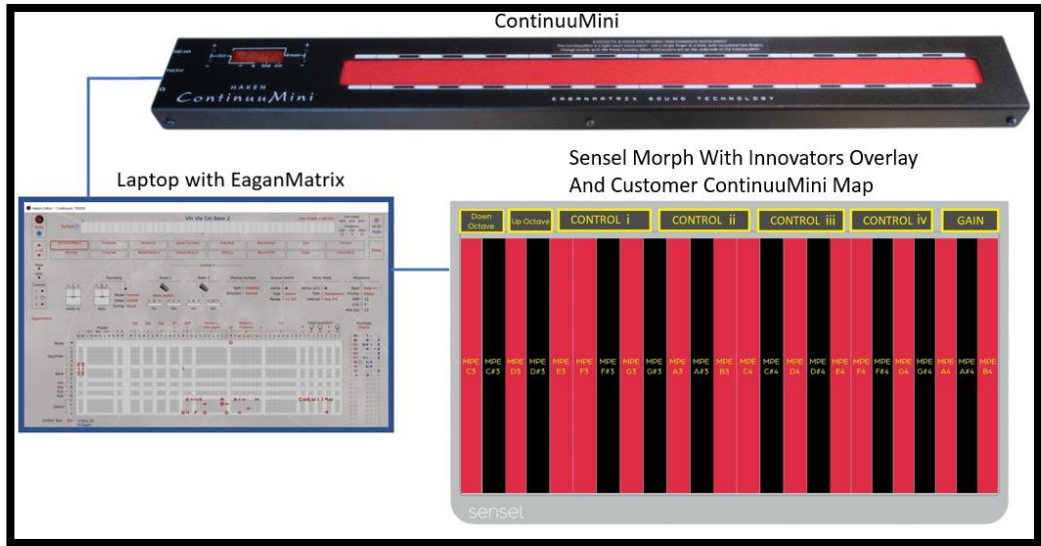


Step 3 - Verify on Continuum Device:

- Set Y/Z= "MPE" for desired preset ("MPE+" will work too)
- Set Z/Bend = 12, 24, 36, 48 or 96 (or whatever you set Roli to – normally you will match Roli to Continuum device)

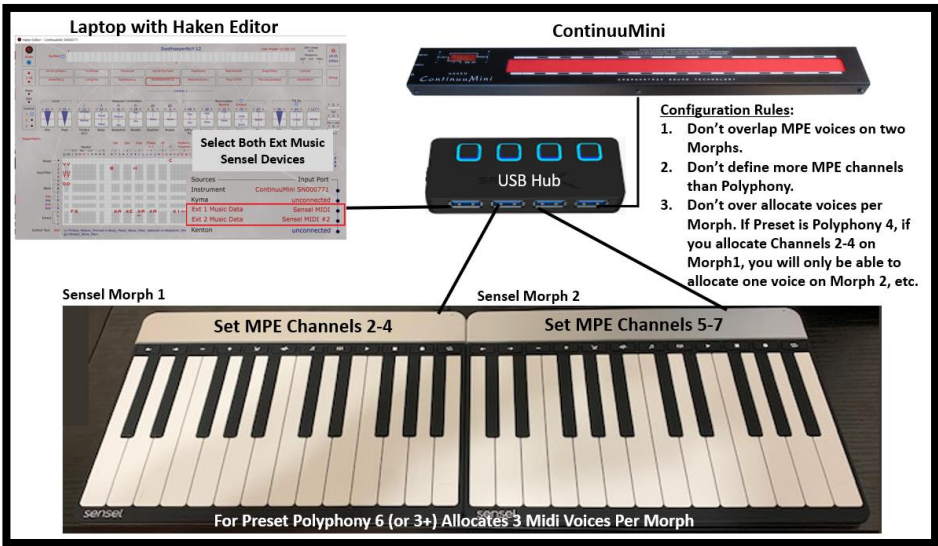
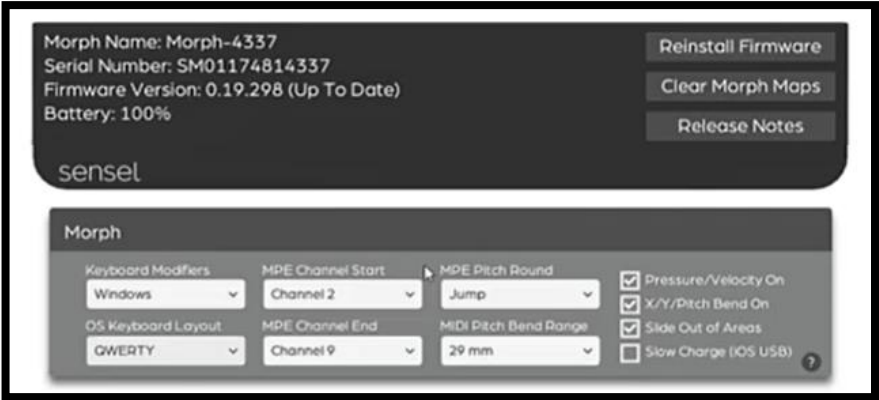
Step 4 – Connect your Roli Block per the desired system diagram to access Continuum Device.

Using Sensel Morph with Continuum USB MPE Devices – Basic MPE Keyboard Setup for ContinuumMini Here



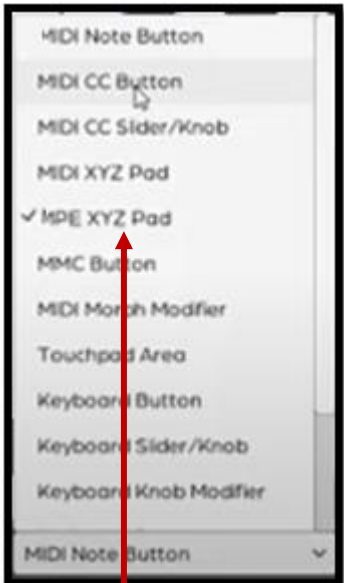
Morph Software Configuration (Top Level MPE Settings):

- 1. Set MPE Channel Start = 2
- 2. Set MPE Channel End = Preset Polyphony +1 (for MPE/MPE+ EaganMatrix Presets)
- 3. Set MPE Pitch Bend mode to Jump or smooth as you desire
- 4. Set Midi Pitch Bend Range = 29mm (mapping to finest bend support)



Morph Software Configuration (Note Configuration):

- 1. Set Notes to desired mappings – Here we'll set MPE XYZ mode to map to Continuum notes (set for each note)
- 2. Set Note mappings for each note. Example here for note Pitch (C3)
- 3. Set Channel Aftertouch to map to Z. Set X to Pitch Bend. Set Y to CC74.
- 4. Set Pitch Bend to map to device. Here 460 mm maps to ContinuumMini PB=48. Match in Haken Editor if needed.
- 5. Set Threshold = 0 (most sensitive pressure detection). Aet Absolute Y and Recenter CC.





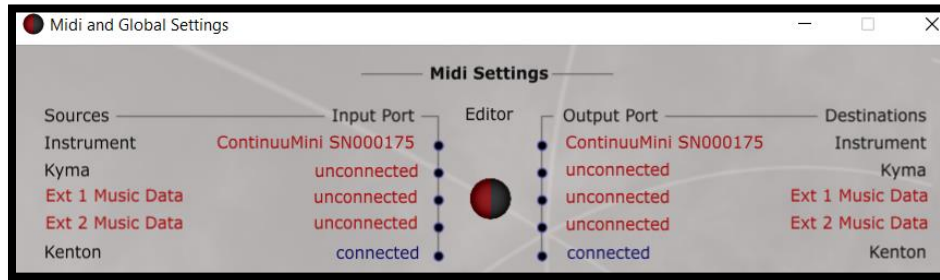
Using Kenton Killamix Mini USB Controller in Editor and Stand-Alone

NOTE: Kenton Killamix Support currently is not working for 9.9x and later firmware.
The issue is currently being addressed

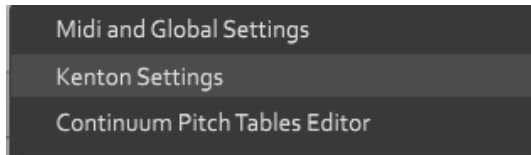
Step 1 – Connect KillaMix to computer or USB hub connected to computer.

Step 2 – Bring up Editor

Step 3 – Go to Midi and Global Settings. Confirm that The Killamix has been automatically detected by the editor.



Step 4 – In Cog Wheel bring up Kenton Settings



Step 5 – Change any mapping you wish if you do not like the defaults. Note this will only change what the current Continuum setting is using. Upon existing the Editor and Continuum device will now respond to the new settings. On repowering the Continuum device, the Killamix defaults will go back into effect as the Continuum internally sets the control assignments.



Playing Moog AniMoog over Wireless Interface from ContinuuMini or Continuum use CME Devices

- CME WIDI Host – to connect USB Midi Host over Bluetooth Wireless
- CME WIDI Jack – to connect TRS-based (or DIN with adapter) Midi devices over Bluetooth Wireless

Playing Animoog Z with ContinuuMini Directly Through CME WIDI Host

iPhone/iPad with Animoog Z

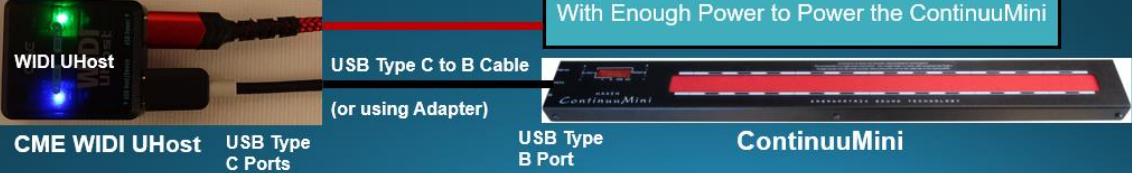


1. Connect WIDI Host to a 5V USB Power Source with ideally at least 500ma (or enough power to power the ContinuuMini)
2. Connect ContinuuMini to the WIDI UHost using a USB C to USB B cable or adapter.
3. Configure ContinuuMini to play on desired Midi channels to play Animoog Z.
4. In Animoog Z Midi Configuration page, select WIDI UHost Bluetooth as input and output (Input to play Animoog Z from ContinuuMini)
5. Make sure BT Input is Selected in Animoog Z Input Section
6. Set Pitch Bend to match Continuum Preset (default 96)
7. Set Animoog Z to MPE mode if playing that way
8. You should now be able to play the Animoog Z App wirelessly through ContinuuMini.

Animoog Z Midi Configuration Panel



Wireless



USB 5V Power Source, USB HUB, etc.
With Enough Power to Power the ContinuuMini

CME WIDI UHost USB Type C Ports

USB Type B Port

ContinuuMini

Playing Animoog Z with Slim Continuum Directly Through BT & CME WIDI Jack

iPhone/iPad with Animoog Z

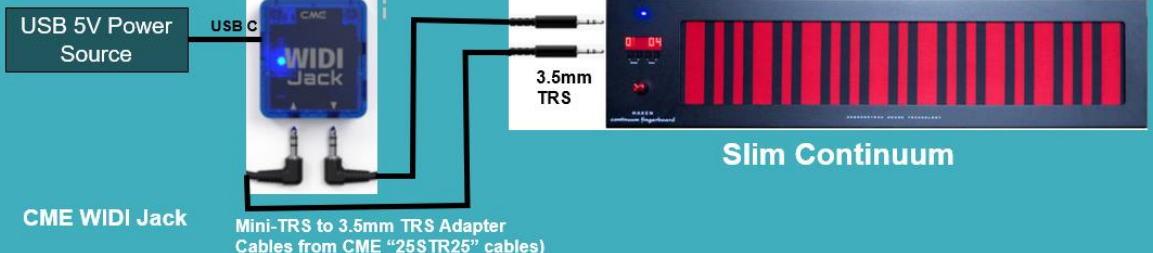


1. Connect WIDI Jack to a 5V USB Power Source (or USB Hub, etc.)
2. Connect the special Mini TRS to 3.5mm TRS adapters to the Midi In/Out ports of Slim Continuum on 3.5mm side and to Midi Jack on the Mini TRS ends.
3. Configure Continuum to play on desired Midi channels to play Animoog Z.
4. In Animoog Z Midi Configuration page, select WIDI Jack Bluetooth as input and output (Input to play Animoog Z from Continuum)
5. Make sure BT Input is Selected in Animoog Z Input Section
6. Set Pitch Bend to match Continuum Preset (default 96)
7. Set Animoog Z to MPE mode if playing that way
8. You should now be able to play the Animoog Z App wirelessly through Slim Continuum.

Animoog Z Midi Configuration Panel



Wireless



USB 5V Power Source

CME WIDI Jack

3.5mm TRS

Slim Continuum

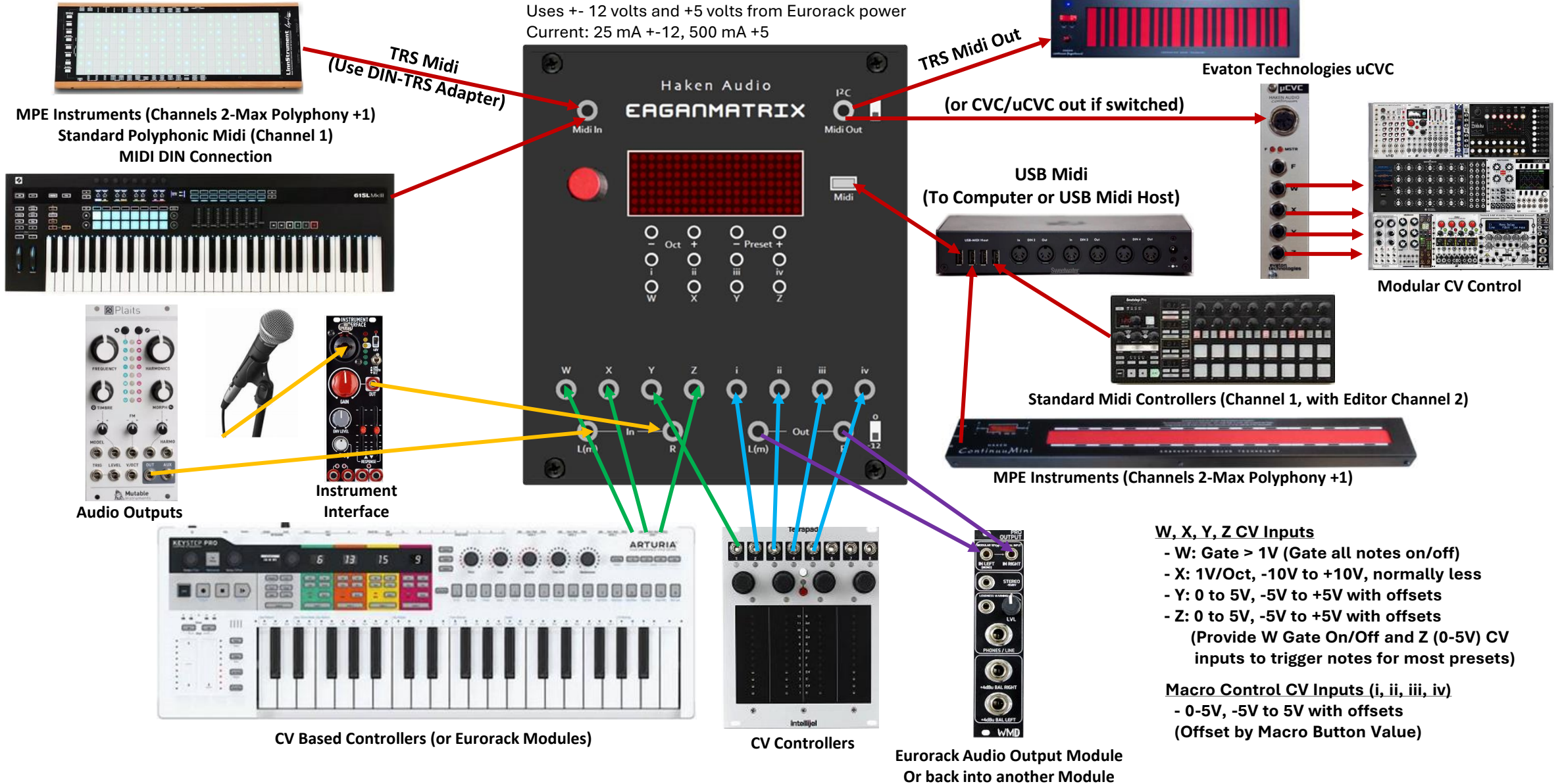
Mini-TRS to 3.5mm TRS Adapter Cables from CME "25STR25" cables)

Note: Animoog is only an example here. Other Apps and wireless Midi interfaces should work as well.

EaganMatrix Eurorack Module Connection Options

Power

Uses +- 12 volts and +5 volts from Eurorack power
Current: 25 mA +-12, 500 mA +5



W, X, Y, Z CV Inputs

- W: Gate > 1V (Gate all notes on/off)
- X: 1V/Oct, -10V to +10V, normally less
- Y: 0 to 5V, -5V to +5V with offsets
- Z: 0 to 5V, -5V to +5V with offsets
(Provide W Gate On/Off and Z (0-5V) CV inputs to trigger notes for most presets)

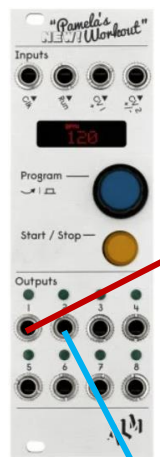
Macro Control CV Inputs (i, ii, iii, iv)

- 0-5V, -5V to 5V with offsets
(Offset by Macro Button Value)

Note: Monophonic through CV Control, Polyphonic through Midi Control (both can be used simultaneously)

Playing the EaganMatrix Eurorack Module With Sequencer and Envelope Generator (One Option)

Clock Source



Passive Multiple (copies clock signals)



Main Clock
Output

Sequencer Input
X Clock

Clock Copy
W Gate

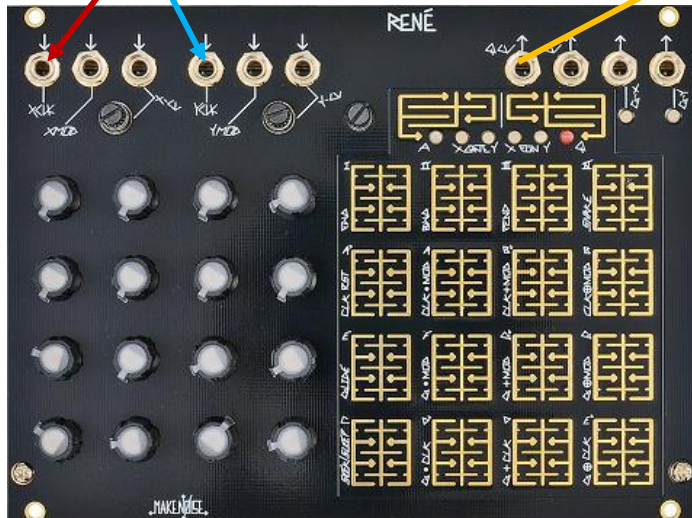
Y Clock



Pitch change
on clock

CV Envelope Output
(Z input)

Envelope Generator
Trigger



Cartesian Sequencer
(takes X and Y clock pulses)



Envelope Generator

Note 1: When using the EaganMatrix module, some presets already have an envelope applied though Shape Generators. Applying an external envelope on that can create some interesting second order envelope effects, but if you do not get the sound you expect, check that the preset is not already applying an envelope.

Note 2: Of course, you could use an active mult instead of passive, but for simple clocking that is not needed. The mult is used to sync all the clocks with the same signal so that the module gates on the same pulse that applies pitch and Z volume shape.

Note 3: For most presets you will Gate with W ($\geq 1V$) while sending a CV (0-5V) volume/expression shape to Z (which could be a gate copy for simplest output though not very musical) and sending a 1V/Oct pitch to the X input typically synced with the W gate. For presets that use Y you can send a separate 0-5V input to control that from the CV output of your choice or press the Y button and use the rotary to manually change Y in real time as you can also do with any of the four Bespoke Controls (i, ii, iii and iv – noting v and vi are also available for control in the Menu System).

Using the EaganMatrix Eurorack Module as a DSP Processor for Audio Input



Instrument Interface
(Converts Instrument input to an acceptable Eurorack level)



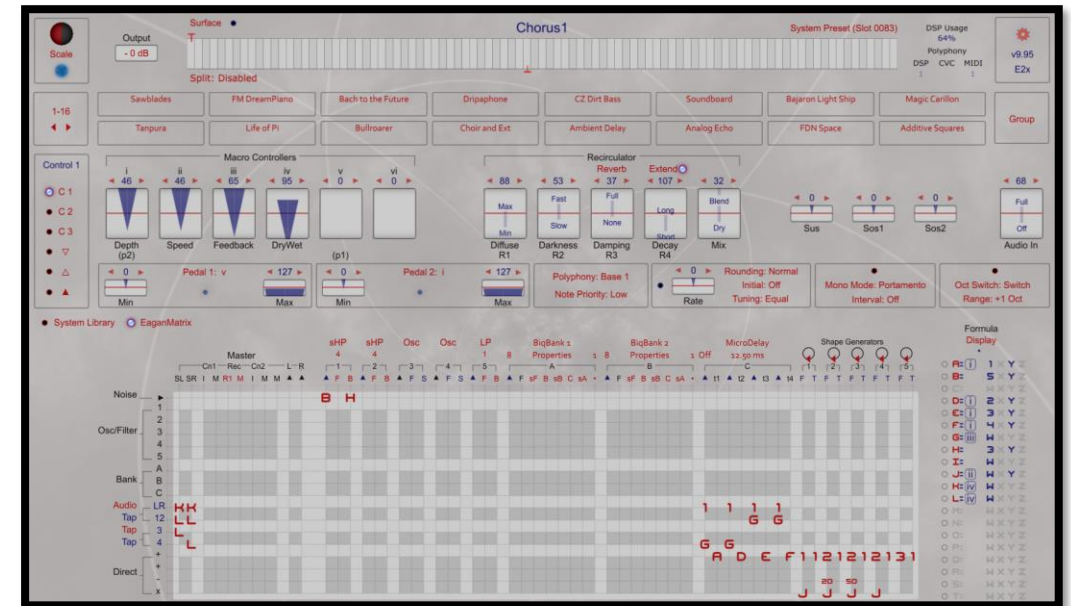
Note 1: Typically, you will take an instrument line level or microphone level input (if not already a Eurorack audio signal that is hotter than line level signals) and run that through a Eurorack Instrument interface (BEFACO used here but there are numerous other options) into one of the EaganMatrix module Audio inputs (or you can use a stereo audio signal as well run to L/R inputs).

Note 2: From here you can load up one of the EaganMatrix module Processor presets to perform digital signal processing of various sorts on the audio input, or code up your own processing preset in the EaganMatrix ("Chorus 1" preset example loaded below).

Note 3: Make sure the Audio Input (AIn) is turned up in the Menu System (or "Audio In" in the Haken Editor or through an external Midi Controller CC19) or no sound will result.



Processor Category: Chorus 1 Preset



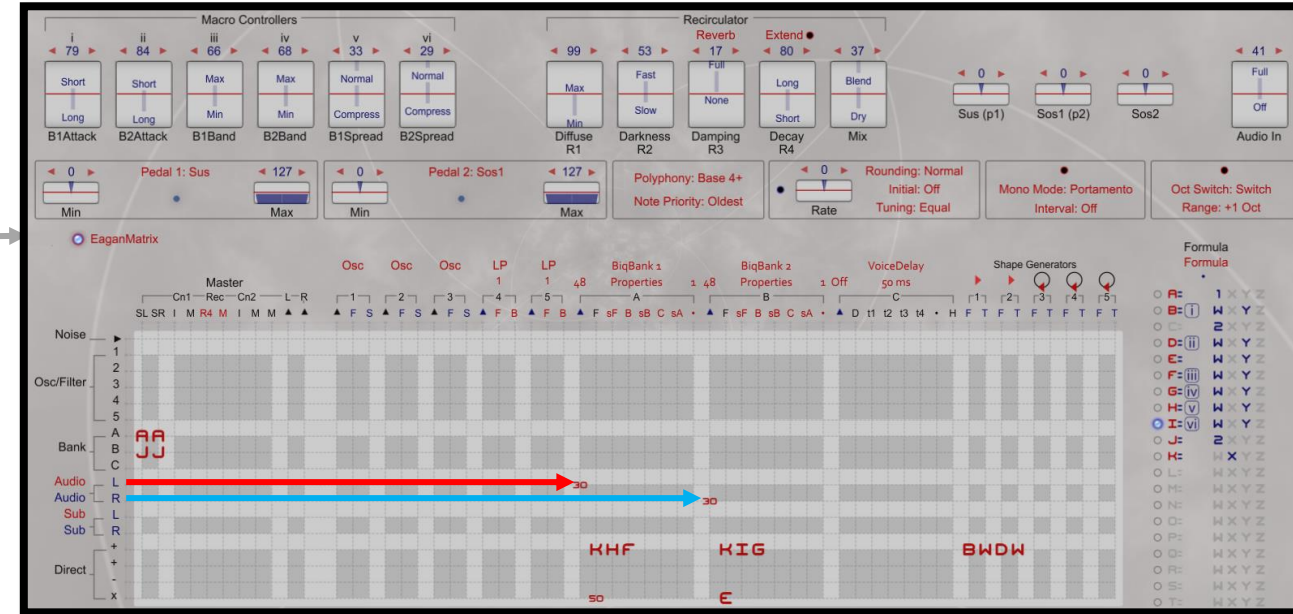
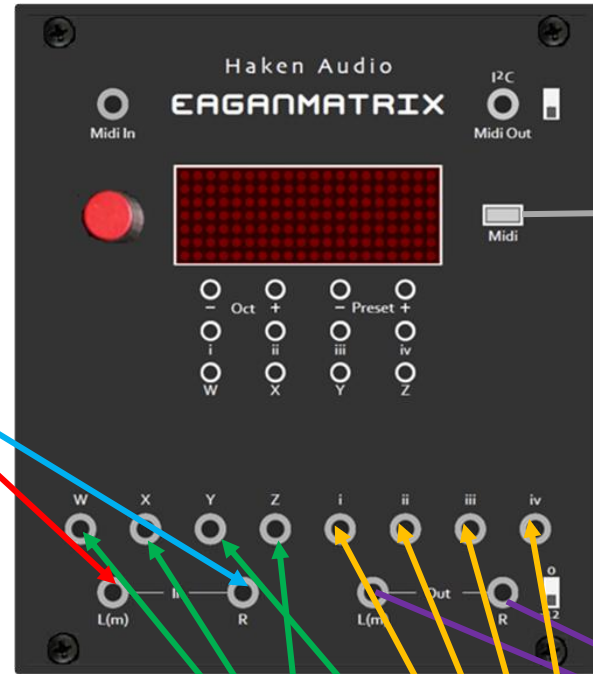
Processing Noise through EaganMatrix Module BiqBanks

Computer Running Haken Editor (Preset Creation & Testing)



Noise Source
(Ideally multiple outputs)

EaganMatrix Module



CV Based Controllers (or Eurorack Modules)



CV Controller
(for Y and Macro Controllers – 0..5V)
(or Buttons and Rotary)



Output Module

W, X, Y, Z CV Inputs

- W: Gate > 1V (Gate all notes on/off)
- X: 1V/Oct, -10V to +10V, normally less
- Y: 0 to 5V, -5V to +5V with offsets
- Z: 0 to 5V, -5V to +5V with offsets
(Provide W Gate On/Off and Z (0-5V) CV inputs to trigger notes for most presets)

Macro Control CV Inputs (i, ii, iii, iv)

- 0-5V, -5V to 5V with offsets
(Offset by Macro Button Value)

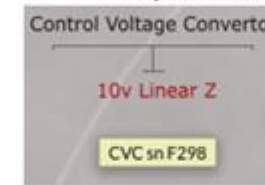
Using Evaton Technologies uCVC with ContinuuMini



Steps

1. Connect uCVC Cable to uCVC and Ped/Ext Input on ContinuuMini (when ordering uCVC specify ContinuuMini or Continuum to get the proper cable).
2. Oct +/- to Enter Menu System and set ContinuuMini i2C setting to 1 (from default 0 pedal setting). Then Oct +/- to Exit Menu System
3. Connect the uCVC outputs as desired
4. Bring up Haken Editor (after uCVC is connected and powered). Verify CVC is connected
5. Set CVC settings as required for the device you are connecting to in CVC settings (in Control Panel 3).
6. Play ContinuuMini and uCVC will output
7. Note: You can chain two uCVCs together to play two voice polyphony. See Evaton Technologies documentation. If you only have one uCVC, it's best to set your EaganMatrix ContinuuMini preset to Polyphony 1 – so you don't have to worry about dropped notes if you mistakenly play polyphonically. If you are not playing along with ContinuuMini analog output, just set an Empty preset to Polyphony 1. If properly connected the Editor will display the CVC as active.

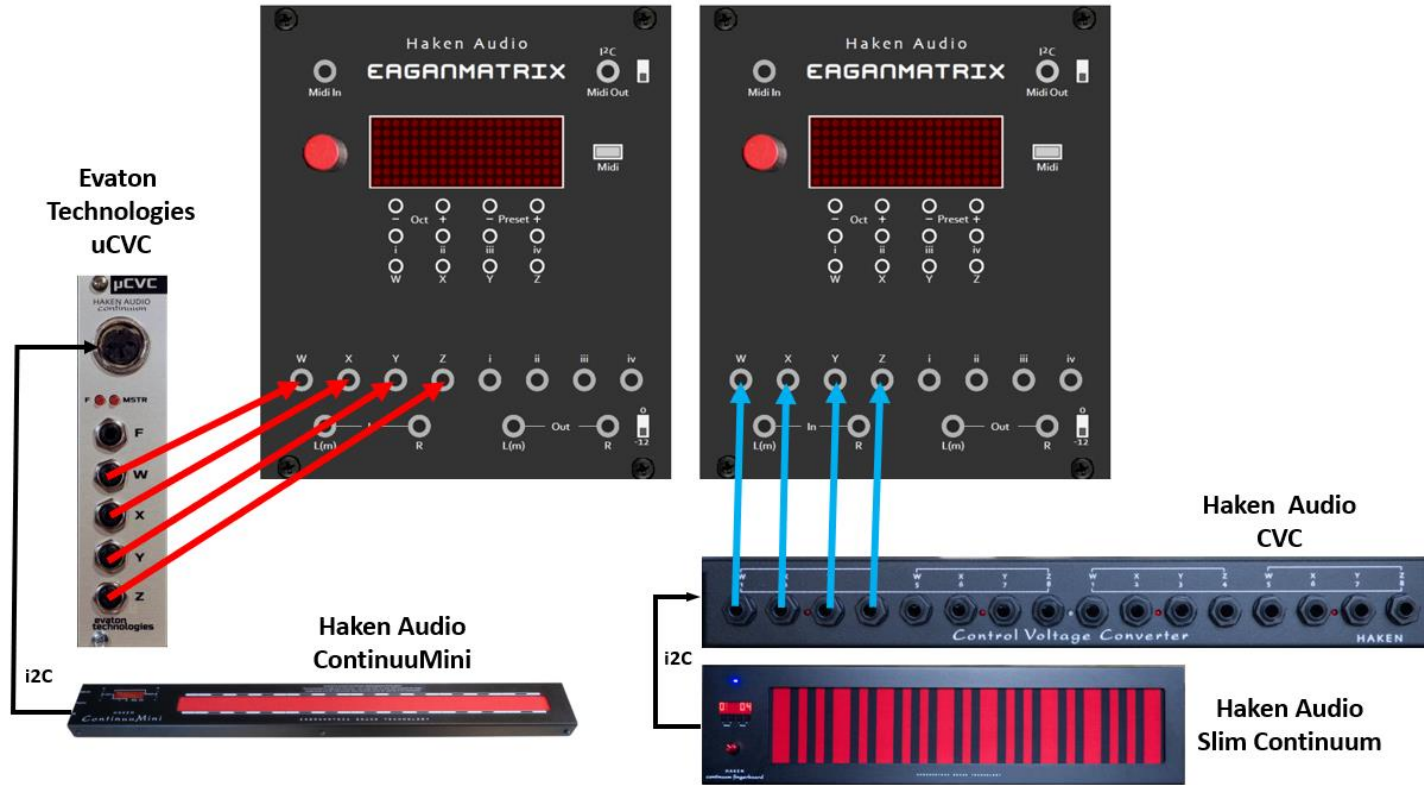
Haken Editor will display CVC=1 in stats if it is detected and CVC device will display if you hover over the CVC display.



EaganMatrix CVC Options (Match to your device)

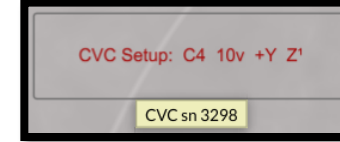
- ✓ 0. CVC Uses 10v (G=10v nn60=0v 1vpoY=0..10v Z=0..10v)
 - 1. 5v C4 (G=10v nn60=0v 1vpoY=-5..5v Z=0..5v)
 - 2. 5v Co (G=5v nn12=0v 1vpoY=-5..5v Z=0..5v)
 - 3. Y Shelf (G=10v nn60=0v 1vpoY=0..10v for Y>.5 Z=0..10v)
 - 4. Voyager (G=10v nn52=0v 1vpoY=-5..5v Z=0..5v)
 - 5. Buchla (G=10v nn10=0v 1.2vpoY=0..10v Z=0..10v)
 - 6. 5v C2 (G=5v nn36=0v 1vpoY=0..5v Z=0..5v)
- ✓ Use Linear Z Values
 - Use Square Z Values
- ✓ Normal Outputs
 - Layer Outputs (when polyphony is 1 or 2)
- ✓ Base Voice 1
 - 2
 - 3
 - 4

Connecting Evaton Technologies uCVC or Haken Audio CVC to EaganMatrix Module

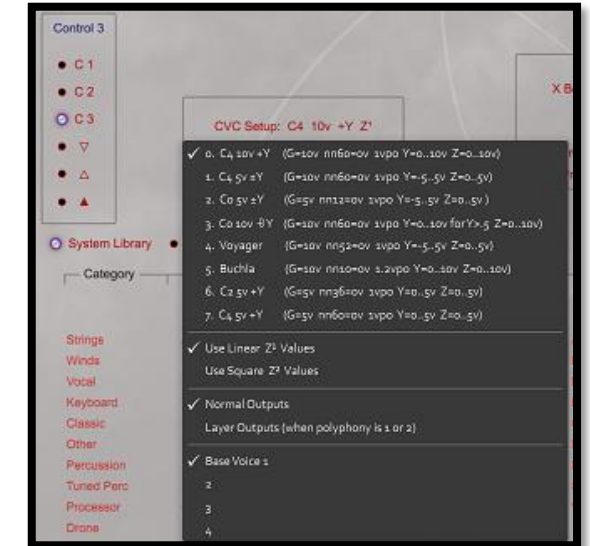


Step 1 – Connect your uCVC or CVC to your Continuum Device (for ContinuuMini set IC2 = 1, Slim Continuum and EaganMatrix module have dedicated connections)

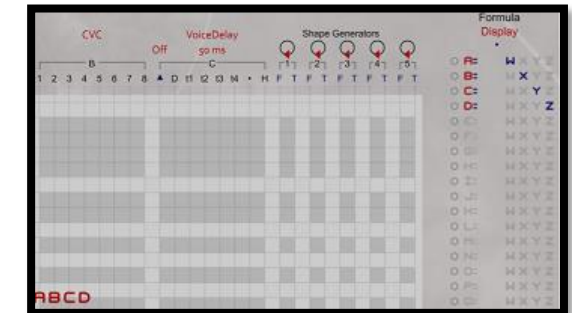
Step 2 – Bring up editor and in Control Panel 3 verify CVC is set. Hovering over CVC section should show you a “CVC sn nnn” message”.



Step 3: Change to desired CV setting to support the device you are connecting to.

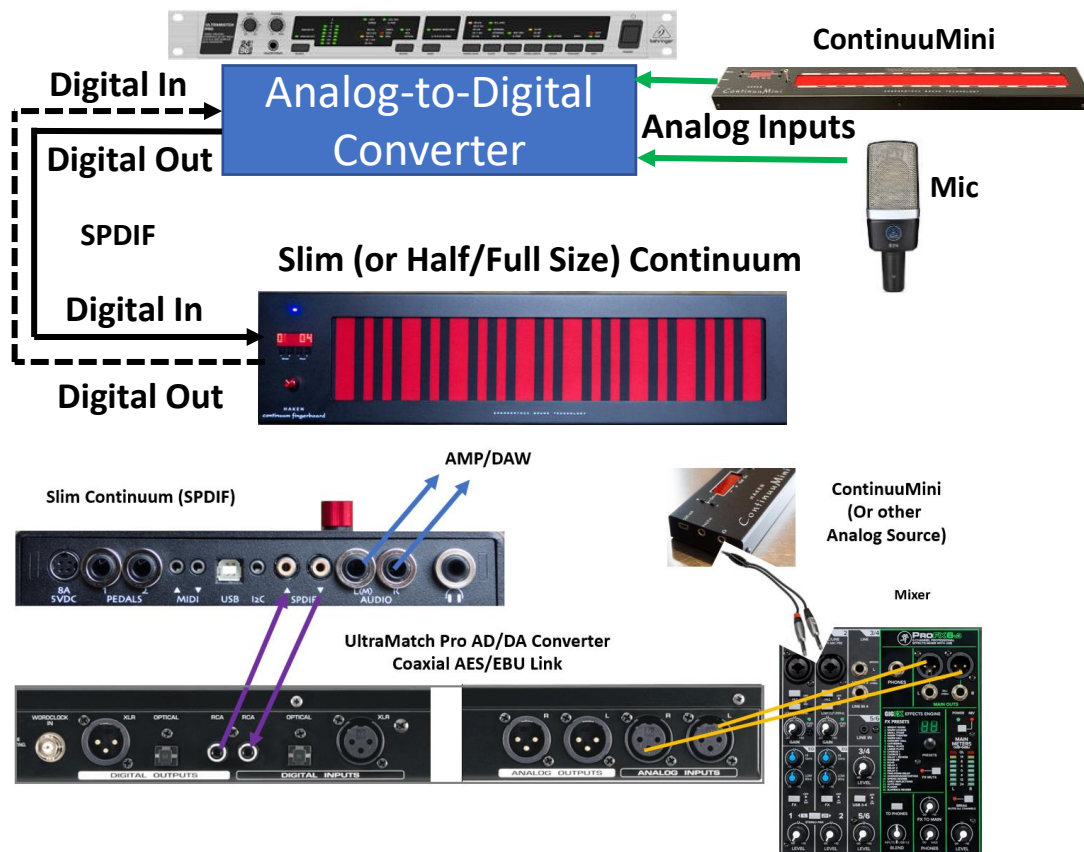


Step 4 – Bring up EaganMatrix if you are programming CVC into a preset (see EaganMatrix Manual or EaganMatrix Preset Cookbook).

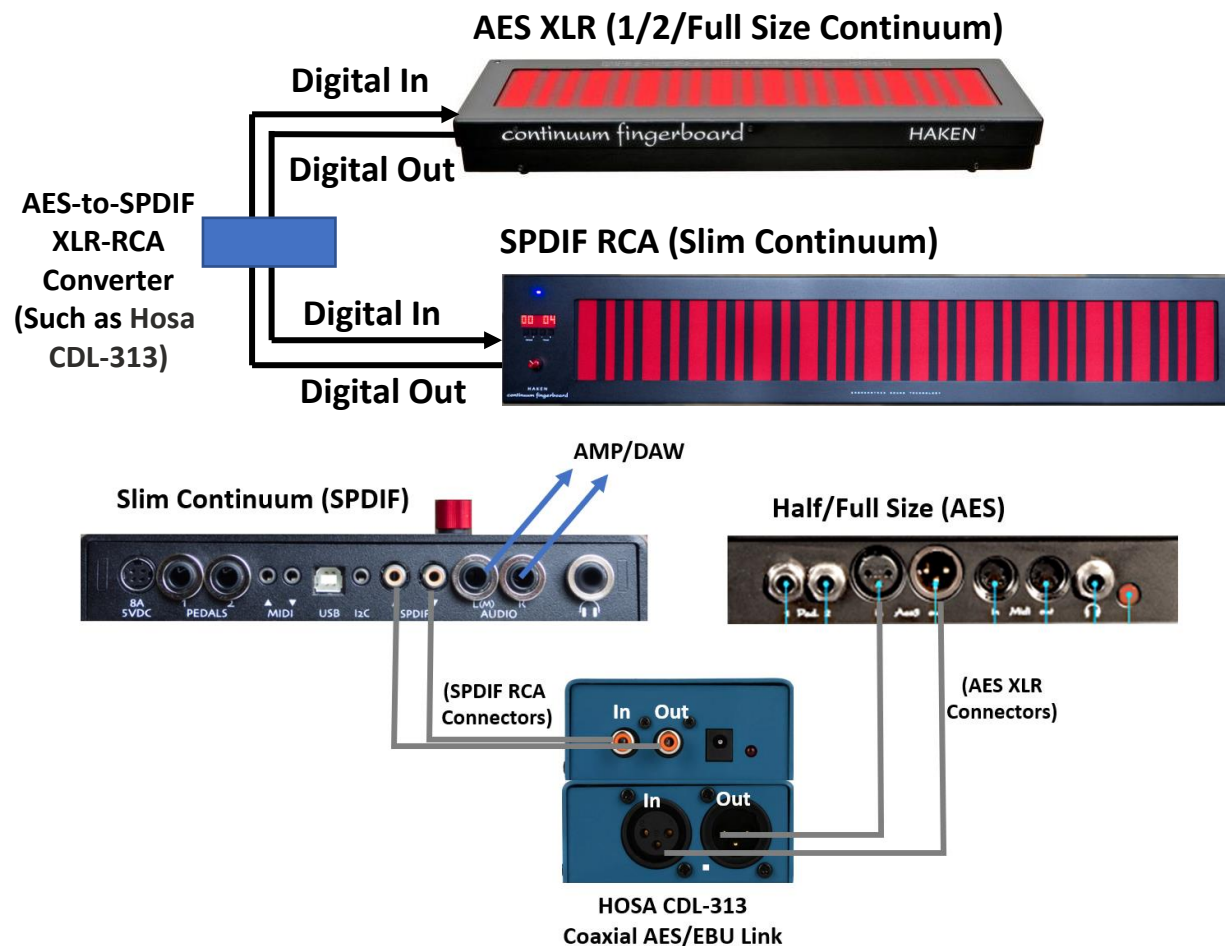


Using AD/DA (AES/SPDIF) Converters with Continuums for Digital I/O

External Analog Audio to Continuum Digital Processing

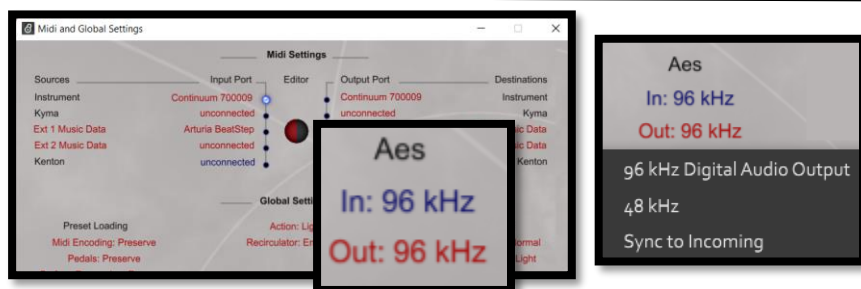


Continuum-to-Continuum Digital Processing



Verifying Continuum AES/SPDIF Digital Connections

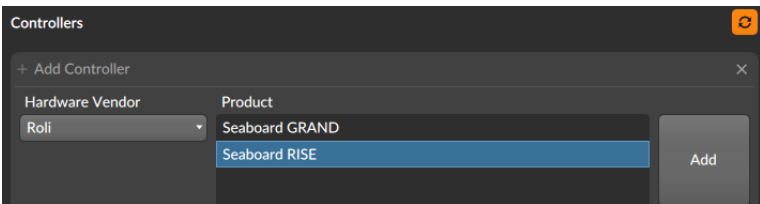
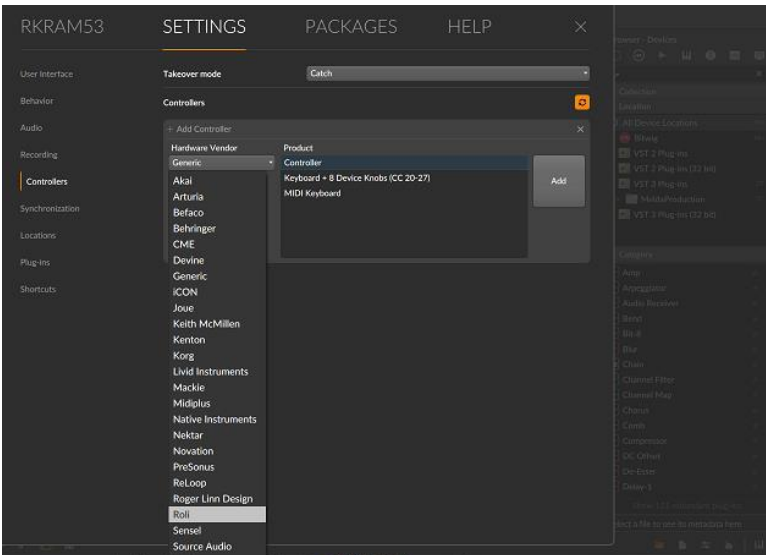
1. Make sure your Digital I/O connections are correct for inputs/outputs
2. "In Midi and Global Settings" you should have AES Input and output frequencies matching
3. If not matching set appropriately or set to Sync to Incoming signal (best if you can match frequencies)
4. Set your converter A/D mode if it has setting
5. For SPDIF or AES/EBU as you find connects and set bit rate (24 bits seems to work)



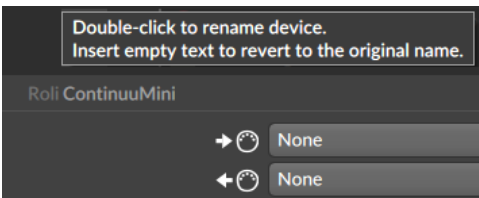
Software Interfacing to Continuum Devices

Step 1 – Connect your Continuum device using USB cable or USB interface (for continuum) and bring up Bitwig.

Step 2 – Go to File/Settings/Controllers and Add a Roli Rise Controller from Hardware vendor Pull down list (there is no Haken selection, this will be close to the settings required).

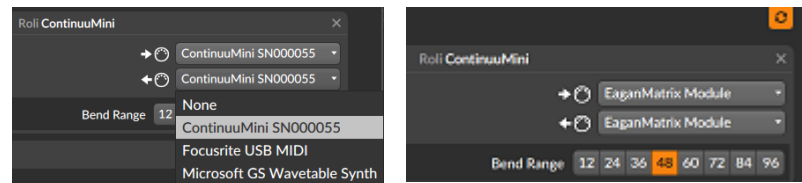


Step 3 – Rename Device to ContinuumMini (or Continuum or EaganMatrix Module). Will show Roli too, but easiest to set this way).

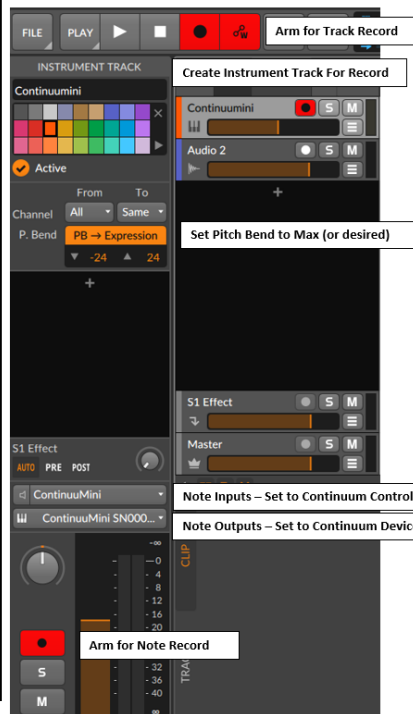
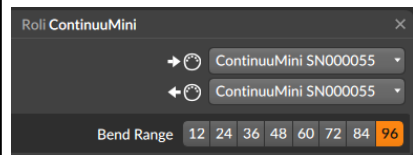


Bitwig DAW Setup For Continuum Device

Step 4 – Set Midi input and output to the Haken Audio USB device (ContinuumMini here – or perhaps EaganMatrix Module if that was used instead, etc.). Note: On Windows you likely will not be able to bring up the editor and run the DAW as they both will try and grab the USB device, first one to get it will block the other from getting it). Also, if you get Midi loop messages on your Continuum device, only set input (for DAW record) or output (for DAW play) here, not both.



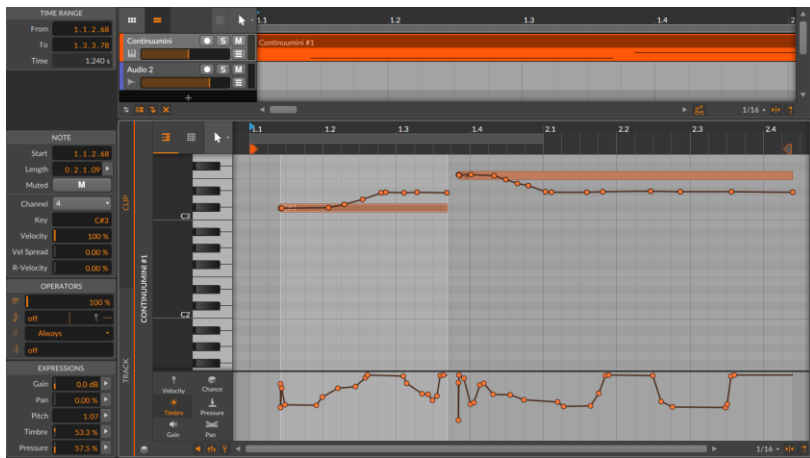
Step 5 – Set Bend Range to match your Continuum preset (normally 96)



Step 6 – To record,

- Create and name your instrument track.
- Set Channel From: All, To: Same. This will make sure all MPE midi Channels are recorded, and you don't have to try and map channels to Polyphony.
- Set Max Pitch Bend and adjust in Controller page if required.
- Set input device to ContinuumMini (or the Controller name you created).
- Arm to Record.
- Press Play arrow to record and Arrow again or Square Stop recording box.
- Note: You may have to limit the amount of data output through pitch bend by increasing rounding as it creates a very large data stream that Bitwig may not handle well

Step 7 – Display and Edit MPE Midi data as you see fit. Pitch bend should be recorded, Timbre and Channel Pressure as well (you can bring up Edit Note Expressions and Micro-Pitch Editing functions). Remember that Continuum will output constant Velocity = 127 unless you change that to Dynamic (but Channel Pressure is your real volume control).

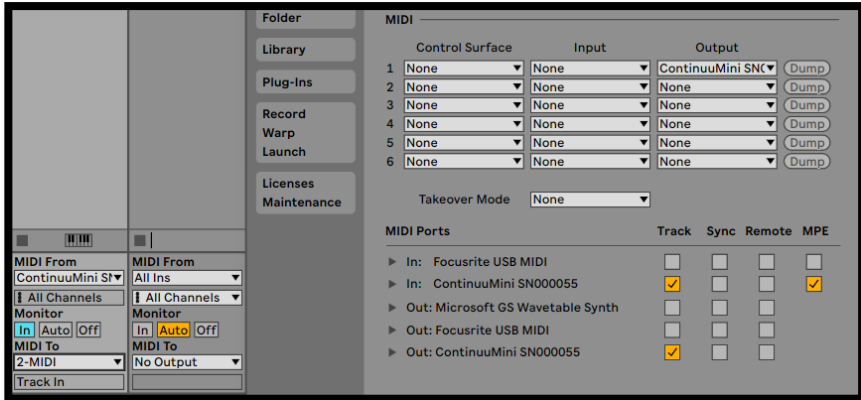


Step 8 – Playback to Continuum Output Note device set as desired Note: May be an issue where it does not play notes back unless set to Channel 1 (need to investigate).

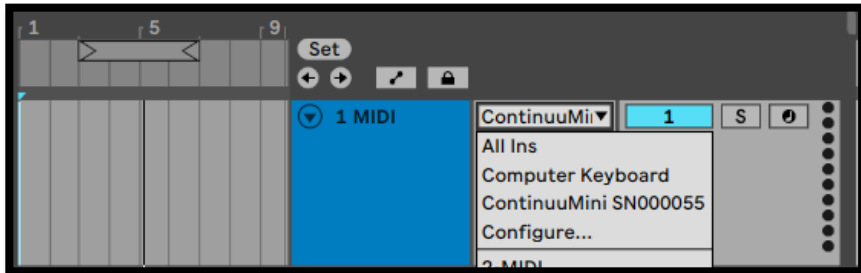
Ableton Live 11 DAW Setup For Continuum Device

Step 1 – Connect your Continuum device using USB cable or USB interface (for continuum) and bring up Ableton Live

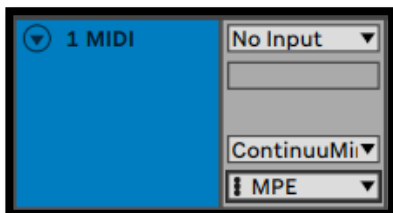
Step 2 – Set Input and output (Midi From, Midi To) devices to ContinuuMini (or other) Continuum device



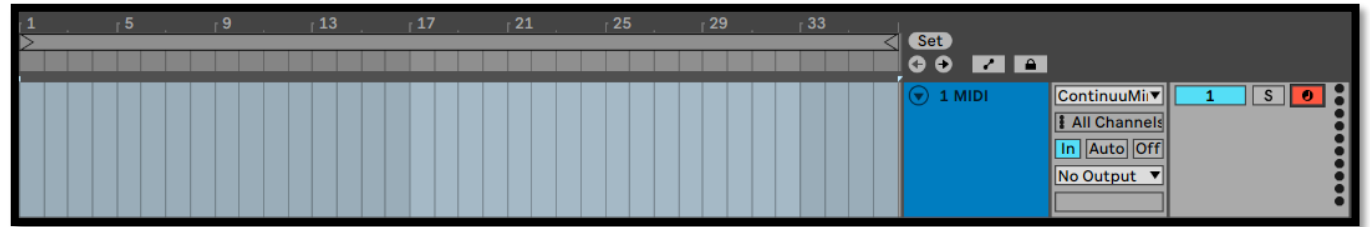
Step 3– In the track you want to record set any specific Input and Output devices. Suggest if you want to record ContinuuMini (etc.) set Input to that device with All Channels set for input. If Setting Output device set to ContinuuMini (etc.) device with MPE option.



Note: Do not set both Input and Output device. If setting Input, Out device should be “No Output”. If setting Output, Input device should be set “No Input”. Otherwise, you will get an ERR! Midi Loop error on Haken Continuum device.



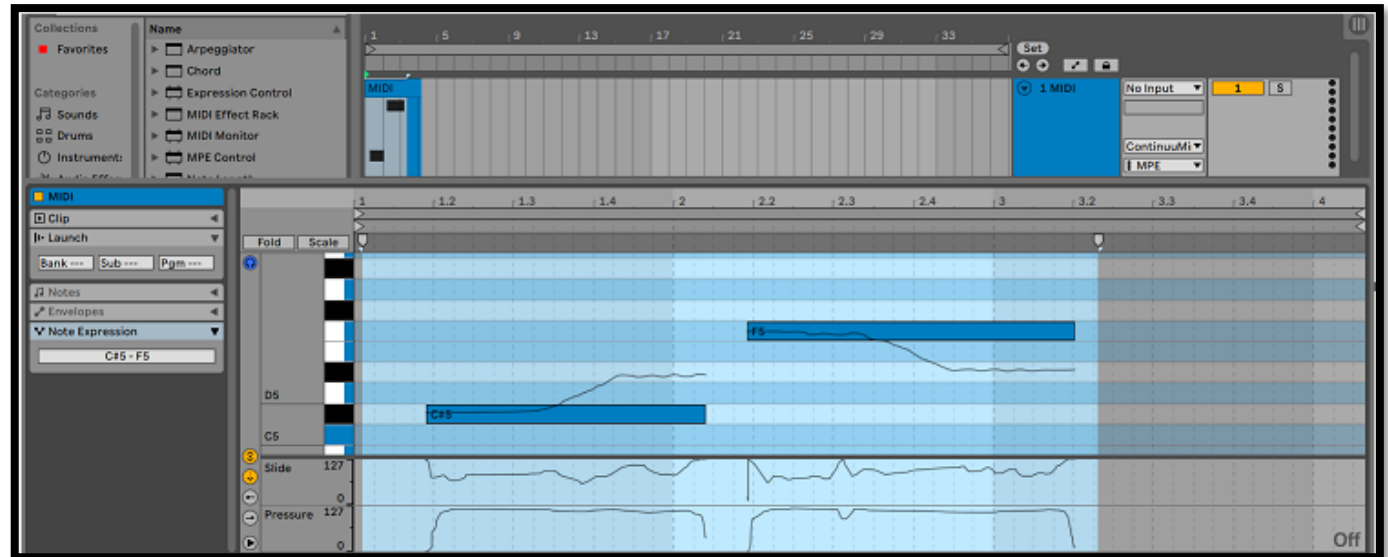
Step 4– Set your range and Arm your track for recording.



Step 5– Press record control to start recording ContinuuMini and Stop control to end.



Step 6– Edit ad you desire, selecting Note Expression option to see your Pitch Bend, Channel Aftertouch (Pressure) and CC74 (Slide) data. Velocity data will default to all notes = 127 unless you set Dynamic Velocity for the preset in the Haken Editor.



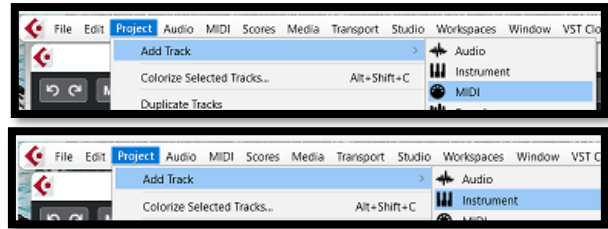
Step 7– Set No Input and then set Output to ContinuuMini device and MPE mode as notes in step 3 to play back to the Continuum device..

Cubase 11 DAW Setup For Continuum Device MPE Record and Playback to Continuum or Plugin

Step 1

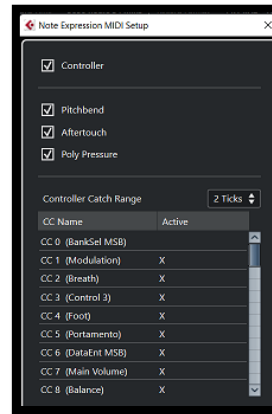
- In the global track controls area of the track list, click **Add Track**.
- Click **Instrument** if you want to play back to a Plugin or click **Midi** if you want to play back to the Continuum.
- If **Instrument**, open the **Instrument** pop-up menu and select your desired plugin (i.e. **HALion Sonic SE**)
- Click **Add Track**.

The Instrument or Midi track is added to the track list, and the control panel of the selected VST instrument is opened if you selected an Instrument plugin.



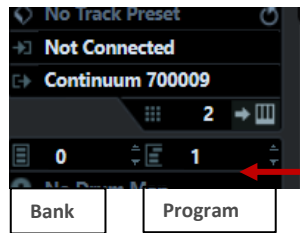
Step 2

Make sure all your desired CCs, Pitchbend and Aftertouch are activated in the Midi/Note Expression menu.



Step 3

Set your Input Device to the Continuum, ContinuuMini, EaganMatrix Module device. Note if you are going to playback to the same Continuum device, do not set the output as that Continuum device or you will create a Midi loop. In that case set the output to Not Connected until you are ready to play back (step 7). If you are using a Instrument Track and Plugin you can simply set your output type to the plugin device or other Midi output device if it is not the Continuum.

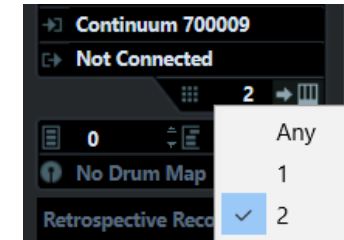


Step 7

If playing back to Continuum in 9.0x and greater you can set a Bank and Program Selectors to desired preset (see Beatstep config slide for Bank/Program options)

Step 4

On Windows make sure you do not have the Haken Editor or some other program connected to the Continuum USB device or Midi input will not be recorded. Cubase says set input channel to anything except Any, but Any seems to work too. Experiment there. Try Channel 2.



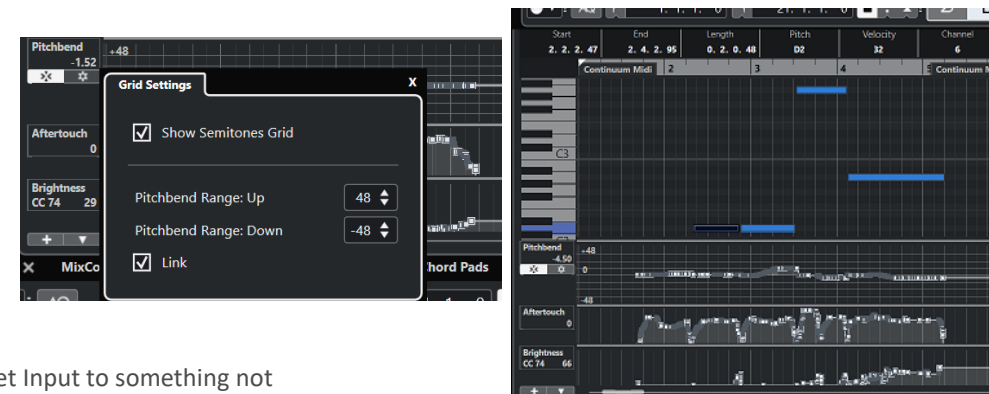
Step 5

Set your record range, ARM your track for record and record Midi as you normally would from any other Midi Device



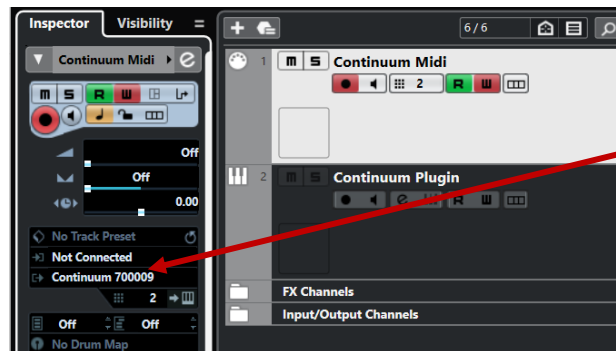
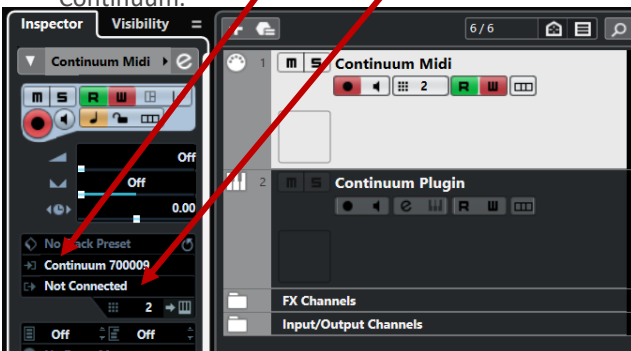
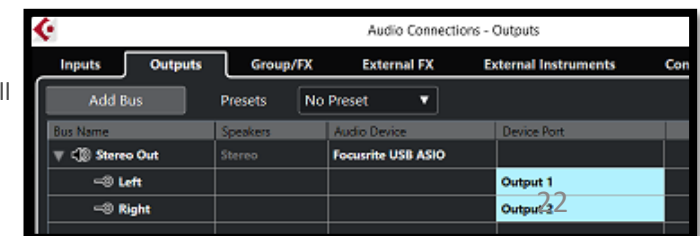
Step 6

Edit as you normally would. Display your Note expression controller lanes for Pitchbend, Aftertouch and CC74/Y. In Set PB range per your preset.



Step 8

For playback, set Input to something not the Continuum device if you are playing back to the Continuum (i.e. **Not Connected**). Set desired output device (plugin or Continuum, etc.). Make sure all your Audio Bus is connected to the desired output devices for playback of audio.

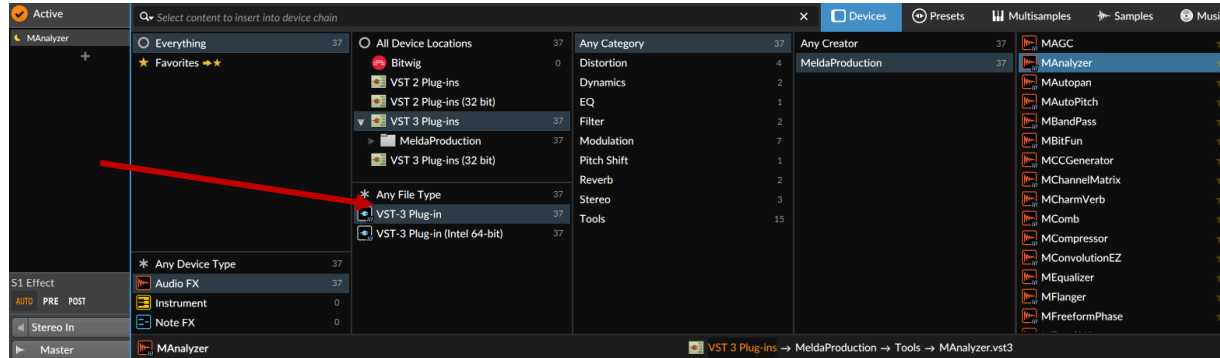


Running VST Plugins (Scopes) with DAWs – Bitwig Example with Melda Plugins on Audio Output

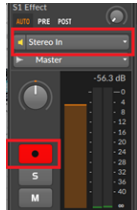
Step 1 – Select a track compatible with the Plugin you want to attach (here an Audio plugin)



Step 2 – Select Add Plugin function for your DAW (here +). And select type of plugin and plugin itself (here VST3, Melda Productions, MAnalyzer)



Step 3 – Make sure track is armed to accept input for your plugin(here Stereo Input)



Step 4 – Select the control that brings up the plugin

